

A proof of the conjectured recurrence for OEIS A243585

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Abstract

OEIS A243585 carries a conjectured linear recurrence with polynomial coefficients, of order 2. The entry gives no generating function, but it does give the sequence as a single hypergeometric sum. We prove the conjecture without assuming any generating function: creative telescoping produces, from the summand alone, an operator of order 2 that annihilates the sequence, together with a rational certificate that is checked as an exact identity rather than trusted. The conjectured operator is then shown to be a left multiple of it by division in the Ore algebra $\mathbb{Q}(n)[N]$. That last step is what the conjecture needs: the two operators have the same order but are not proportional, so the conjecture still has to be derived from what telescoping gives.

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1 The sequence and the conjecture

OEIS A243585 is “Expansion of $x \cdot \log'(C(C(x)-1)-1)$, $C(x) = (1-\sqrt{1-4x})/(2x)$ ”. It has offset 0 and begins

$$a(0), \dots, a(7) = 1, 4, 20, 106, 580, 3244, 18446, 106250, \dots$$

The entry states, not as a conjecture,

$$a(n) = \text{Sum}_{k=0..n} \text{binomial}(2k, k) \cdot \text{binomial}(2n, n-k).$$

so that, writing $F(n, k)$ for the summand,

$$a(n) = \sum_{k=0}^n F(n, k), \quad F(n, k) = \binom{2k}{k} \binom{2n}{-k+n}. \quad (1)$$

This is the only input the proof takes from the entry, and it was checked against the entry’s own DATA before use.

The entry separately carries the following comment:

Conjecture D-finite with recurrence: $2 \cdot n \cdot (2 \cdot n - 1) \cdot (3 \cdot n - 5) \cdot a(n) + (-123 \cdot n^3 + 328 \cdot n^2 - 249 \cdot n + 60) \cdot a(n-1) + 50 \cdot (n-1) \cdot (2 \cdot n - 3) \cdot (3 \cdot n - 2) \cdot a(n-2) = 0$. - *R. J. Mathar*, Jun 14 2016

As of the “Last modified” line on the live entry (2025-11-05, revision 33) the statement is still recorded as a conjecture, and no proof appears among the entry’s links, formulas or programs.

2 Telescoping the summand

Let N denote the shift in n , $(Nf)(n) = f(n+1)$, and Δ_k the forward difference in k , $(\Delta_k g)(k) = g(k+1) - g(k)$.

Lemma 1. Suppose there are rational functions $\sigma_0, \dots, \sigma_r$ of n , not all zero, and a rational function $R(n, k)$ such that

$$\sum_{i=0}^r \sigma_i(n) F(n+i, k) = \Delta_k [R(n, k) F(n, k)]. \quad (2)$$

Suppose further that $F(n, k) = 0$ for every k outside the range of summation in (1). Then $\sum_i \sigma_i(n) a(n+i) = 0$.

Proof. Because $F(n, k)$ vanishes off the stated range, the sum in (1) may be taken over all $k \in \mathbb{Z}$ without changing it, and the same holds for each $F(n+i, k)$. Summing (2) over all k , the left side becomes $\sum_i \sigma_i(n) a(n+i)$ and the right side telescopes: the partial sums of a forward difference collapse to the boundary values, which are 0 because $R(n, k) F(n, k)$ vanishes for $|k|$ large. Hence the left side is 0. $\square$

The point of Lemma 1 is that (2) is an identity between rational functions once both sides are divided by $F(n, k)$: the quotients $F(n+i, k)/F(n, k)$ and $F(n, k+1)/F(n, k)$ are rational because F is hypergeometric in both variables. So a proposed (σ, R) can be *checked*, exactly, whatever produced it.

For A243585 such a pair exists with $r = 2$:

$$\sigma_0(n) = -5, \quad \sigma_1(n) = \frac{123n^3 + 410n^2 + 413n + 110}{60n^3 + 170n^2 + 150n + 40}, \quad \sigma_2(n) = \frac{-6n^3 - 23n^2 - 25n - 6}{30n^3 + 85n^2 + 75n + 20}$$

and certificate

$$R(n, k) = \frac{15k^3n + 20k^3 + 9k^2n^2 - 16k^2 - 18kn^2 - 46kn - 30k}{15k^3n + 20k^3 - 15k^2n^2 - 50k^2n - 40k^2 - 15kn^3 - 50kn^2 - 55kn - 20k + 15n^4 + 80n^3 + 155n^2 + 1}$$

Dividing (2) by $F(n, k)$ and cancelling gives 0 identically, so Lemma 1 applies and

$$L = \sum_{i=0}^2 \sigma_i(n) N^i \quad (3)$$

annihilates a .

3 From the derived recurrence to the conjectured one

The operator (3) is not the conjectured one. It has the same order as the conjectured one but is not a multiple of it by a constant. The two are connected by division in the Ore algebra $\mathcal{A} = \mathbb{Q}(n)[N]$, where multiplication obeys $Nq(n) = q(n+1)N$.

Lemma 2. If $C = QL$ in $\mathcal{A}$ and $L(a) = 0$, then $C(a) = 0$ at every n where the coefficients of Q are defined.

Proof. $C(a) = (QL)(a) = Q(L(a)) = Q(0) = 0$. The coefficients of Q are rational functions of n , so the step is valid away from their poles. $\square$

Writing the conjectured recurrence as an operator C and dividing by L leaves remainder 0, with quotient

$$Q = (-60n^3 - 170n^2 - 150n - 40).$$

No coefficient of Q has an integer pole, so Lemma 2 applies at every index.

Theorem 3. The conjectured recurrence

$$2n(2n-1)(3n-5)a(n) + (-123n^3 + 328n^2 - 249n + 60)a(n-1) + 50(n-1)(2n-3)(3n-2)a(n-2) = 0$$

holds for every $n \geq 2$.

Proof. By Lemma 1 the operator (3) annihilates a , and by Lemma 2 applied to $C = QL$ so does the conjectured operator. No coefficient of Q has an integer pole. $\square$

4 Verification

Every number above is an output of a script written separately from this note, and four independent checks were run.

First, the certificate was verified, not trusted: (2) was divided by $F(n, k)$ and the difference of the two sides cancelled to 0 as a rational function of n and k .

Second, the vanishing of $F(n, k)$ off the range of summation — the hypothesis Lemma 1 needs, and the one that is easy to assume without looking — was checked directly, on concrete integers on both sides of the range.

Third, the factorisation was confirmed by multiplying back: QL was expanded in $\mathcal{A}$ and its coefficients cancelled against those of C to zero.

Fourth, the conjecture itself was evaluated on the published terms, in exact integer arithmetic and without reference to any operator: for each n in range the sum $\sum_i p_i(n)a(n-i)$ was formed from the entry's own values and found to vanish, for all 19 values of n from $n = 2$ upward. The formula (1) was likewise checked term by term against the DATA section before being used.

References

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